

Mapping intra-urban social deprivation from Sentinel imagery using only municipal aggregate labels

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Abstract

Most countries publish social statistics as administrative aggregates; almost none publish them at neighborhood level. We measure how much neighborhood detail aggregates and free satellite imagery jointly recover. Trained only on five-level deprivation aggregates of Mexican municipalities, with no spatial annotation, and scored against fully held-out census-tract grades, the map reaches a within-municipality rank correlation of $\rho = 0.20$ on fourteen test cities opened once, with frozen models: 84% of what the same features attain there under full tract supervision (0.23; 90% on the validation cities used for selection). More aggregates keep buying map, state aggregates forfeit part of it, one national figure erases it, and optical content matters where resolution does not. The maps close 31–66% of the aggregate-to-census targeting gap on test, replicate against an independent institution’s index, beat the incumbent wealth index where selection occurred and match it on test, and order Bogotá’s block strata without retraining. Against free global grids the margin is resolvable only for population; nighttime lights order tracts inside a municipality as well as the maps do. Attention-based multiple-instance learning fails here for a measurable, structural reason, and ensembling over seeds yields no usable uncertainty layer: an honest 90% grade interval spans 1.7 of the five levels.

1 Introduction

Census-quality social data is expensive, and most of the world sees it roughly once a decade. In between, and below the level the tables are published at, policy relies on aggregates: municipal indices and district averages. A rich literature has shown that satellite imagery plus machine learning predicts such indicators across space [1–4], and derived global products already steer programme targeting [3, 5]. Predicting an aggregate somewhere else and disaggregating it in place are different tasks, and the second is the one targeting decisions require: a mayor already knows the municipality is poor but lacks a map of which neighborhoods are poorest.

We ask the disaggregation question quantitatively: how much neighborhood detail can be recovered from a country’s published aggregates combined with free imagery? We frame the answer as an exchange rate between the number and granularity of published aggregates and the within-municipality detail they recover, and we measure it in a country that publishes ground truth at both levels. Mexico publishes its Social Deprivation Grade (GRS) both as municipal

aggregates and for 61,430 urban census tracts (AGEB) [6]; we train only on the former and score maps against the latter, held out entirely, in cities the model never saw (Fig. 1).

Three design choices constrain the measurement. Supervision is weak: a model sees a bag of image tokens for a municipality and one vector, the population shares at each grade threshold, with no spatial annotation of any kind. Every comparison is scored *within* municipalities, because ordering tracts across cities is mostly solved by knowing which city is poor (pooled correlations run near 0.46 against 0.23 within, at the same operating point). And an oracle trained directly on the held-out tract labels bounds what the frozen representation supports, so weak supervision is always reported as a fraction of a measured upper bound.

2 Data

2.1 Imagery

Sentinel-2 L2A and Sentinel-1 RTC annual median composites for 2020 (10 m grid, windowed COG reads from Microsoft Planetary Computer), for 424 boxes covering the 138 cities of the national set and the 213-municipality expansion; city boxes are conurbations and typically span several municipalities.

2.2 Labels

CONEVAL GRS 2020: municipal-level population shares at each of four grade thresholds supervise training; AGEB-level grades are used only for evaluation and for the oracle. Auxiliary evaluation sources: CONAPO IMU 2020 [7], ESA WorldCover 2020 [14], Meta RWI [3], IBGE AGSN 2019 [18], Bogotá’s district stratification [19].

3 Methods

3.1 Bags and instances

A bag is a municipality (771 after exclusions); instances are 16-pixel tokens (160 m) encoded by a frozen DOFA foundation model [13], both sensors fused per token. Municipalities contributing ground to any held-out city are excluded from the expansion; an earlier version leaked nine validation municipalities this way, detectable only at bag level, and every affected number was re-measured.

3.2 Model

A two-layer head on standardized frozen features predicts four cumulative-grade probabilities per token; the bag prediction is the mean of token predictions, fit with binary cross-entropy; each token’s hidden representation is concatenated with the mean of its 8 grid neighbours (480 m), computed city-wide at inference.

3.3 Protocol

110/14/14 cities for train/validation/test, stratified by size and deprivation, no city split across sets. The test cities were opened exactly once, after all development ended: every configuration retrains with early stopping anchored to the validation cities as during development (deterministic given the seed), or loads its saved weights, and the test bags are only ever scored under **inference mode**. Nothing was tuned, selected or iterated against test results. Sweeps select on grouped training folds under bag error only; validation cities are scored once per reported configuration; comparisons are paired by municipality over three seeds; every interval on a within-municipality mean is a city-clustered bootstrap (2,000 resamples), and curve error bars are s.d. over seeds.

3.4 Oracle

The same head trained on tract labels, with the same context, bounds the representation ($\rho = 0.251$ with context, 0.202 without).

3.5 Zero-shot transfer

Bogotá and Rio get the same 2020 composite protocol; tokens are scored by the three final seeds with Mexican standardization statistics, deliberately unadapted. Rio’s evaluation is restricted to tokens with WorldCover built fraction ≥ 0.10 .

3.6 Sweep

Fourteen configurations varied one ingredient each (learning rate 5e-5–1e-3, hidden width 64–512, dropout 0–0.5, weight decay, loss, input standardization), then a combination round; standardized inputs with binary cross-entropy won on bag error over five grouped training folds.

3.7 Targeting simulation

For each validation city and budget $q \in \{5, 10, 20, 30\}\%$ of population, tracts are funded in the order given by each allocator (municipal aggregate with ties broken at random over 200 draws, the map’s tract-mean score, or tract truth), with the marginal tract funded fractionally, so the truth ordering upper-bounds both by construction. The outcome is the population living in high or very-high grade tracts among those reached, pooled over cities before the gap ratio is formed.

3.8 RWI pairing

Each AGEB takes the RWI value of the nearest grid point to its centroid; the model is averaged token-to-tract; both are scored on the 2,958 shared AGEB in the 22 municipalities with at least 20 scored tracts and more than one grade, and the interval is a city-clustered bootstrap of the per-municipality difference.

3.9 Global product comparison

Four free global products stand in for what a practitioner would download instead of training anything: GHSL built-up surface (R2023A, 2020 epoch, 100 m), GHSL average net building height

(R2023A, 2018, the only epoch published), GHSL residential population (R2023A, 2020) [15], and the harmonized global nighttime lights of 2020, VIIRS-derived and expressed on the DMSP digital-number scale at 30 arcsec [16]. Nighttime lights come from the harmonized series because the EOG annual composite requires an account; the harmonized layer is the same VIIRS observation on a coarser grid and a compressed scale, which is a handicap for intra-urban ranking and is reported as such. Each product is averaged over the pixels whose centres fall inside a tract, in the product’s own grid so nothing resamples it; a tract smaller than one cell takes the value at a representative interior point, and the count of such fallbacks is reported. All four are used in practice as development proxies, so deprivation is taken as the negative of each, an orientation fixed before any correlation was computed. Scoring is the RWI protocol unchanged.

3.10 Uncertainty and intervals

Uncertainty is read from the three training seeds: a tract’s map value is the mean of its per-seed tract means and its spread is their standard deviation, a quantity computable with no labels. Three things are then asked of that spread. Whether it ranks the errors, as the within-municipality correlation between spread and the absolute distance from an isotonic score-to-grade map fitted on validation. Whether discarding the least confident tracts inside each municipality raises the correlation over what remains; the panel of municipalities is fixed to those that stay above twenty tracts at every point of the sweep, so the curve cannot rise by attrition. And whether an interval can be attached, by split conformal calibrated on the validation cities and evaluated on the test cities, with the absolute isotonic residual as the nonconformity score. Two rules are compared: the marginal quantile, which treats all tracts as exchangeable, and a clustered rule that takes each calibration city’s quantile and then the 90th percentile across cities. Per-city coverage is reported for both.

3.11 MAUP bins

Tracts are binned by tokens contained (1–3, 4–9, 10–19, 20–49, ≥ 50); within-municipality correlations use municipalities with at least ten tracts in the bin.

3.12 Border test

Adjacent token pairs (160 m apart) are classed by what their midpoint crosses; the excess mean absolute score jump of same-grade municipal-border pairs over same-tract pairs is city-bootstrapped; same-grade AGEB-border pairs inside one municipality are the placebo.

3.13 Software and compute

Python 3.12 with PyTorch, rasterio, geopandas and scikit-learn, pinned by the repository’s `uv.lock`; all experiments ran on one Apple-silicon laptop (MPS backend). Seeds 0, 1 and 2 throughout. The seven municipalities excluded from the expansion for insufficient composite depth or persistent acquisition errors (Supplementary Table 3) are the only data exclusions.

3.14 Data and code

The full pipeline, frozen vectors, labels, splits and protocol are released; every figure regenerates from one canonical results file.

4 Results

4.1 Supervision efficiency

Figure 2 shows the measurement on a pool of 771 municipal bags: the 110 training cities plus 213 municipalities added expressly to cover the deprived half of urban Mexico that large-city samples miss (Fig. 1a; municipalities overlapping any held-out city’s ground are excluded, Methods). Every row below was scored once on the fourteen test cities with frozen, validation-selected models after all development ended; validation values are shown beside them as the selection benchmark. **Quantity** (Fig. 2a): fifty aggregates yield $\rho \approx 0.15$ within municipalities; the full national supply yields 0.196 on test against a fully supervised upper bound of 0.233 there (per-point values in Supplementary Table 1), an 84% recovery (validation: 0.227 of 0.251, 90%). Pairing model and oracle by municipality, the city-clustered bootstrap puts the test fraction at 0.85 with a wide interval [0.30, 1.86]; fourteen cities bound the precision, and it is the point estimate, consistent across splits, that the title claims. The test curve reproduces the rise through 400 aggregates almost exactly and flattens at the top: the final gain observed on validation for the full supply is not fully reproduced. **Granularity** (Fig. 2b): relabeling the same bags with state-level aggregates cuts the map from 0.225 to $\rho = 0.152$ on validation and from 0.195 to 0.168 on test; a single national aggregate erases it (0.009 validation, 0.004 test). Recovered map quality falls monotonically with coarser publication on both splits. **Content** (Fig. 2c): block-averaging the optical input to the radar’s 20 m effective resolution costs little on validation (0.183 $\rightarrow$ 0.172) and more on test (0.179 $\rightarrow$ 0.129), while radar sits far below both (0.006 validation, 0.084 test), despite having been the best single sensor at detecting high-deprivation tracts in our feature-based phase-one baseline (per-city AUROC 0.773 against 0.747 optical). Sensor content dominates ground sampling distance, with resolution carrying a real secondary cost on test; a sensor’s prediction accuracy is a poor guide to its map quality.

4.2 Failure mode of attention-based pooling, and an instance-level alternative

The standard weakly supervised localizer, gated-attention multiple-instance learning (ABMIL [8] with the CLAM instance constraint [9]), predicts municipal labels well (quadratic-weighted $\kappa = 0.62$) and produces a map at chance (AUROC 0.463; Fig. 3a; these diagnostics ran on the national-set pool). An entropy penalty that flattens the attention to near-uniform costs no bag AUROC (0.899 $\rightarrow$ 0.895): the bag task is solvable from the plain feature mean, so the attention weights receive no gradient that rewards localization.

Reversing the order of prediction and aggregation removes the failure. Predicting every token and averaging the predictions (deep learning from label proportions [10–12]) ties the training signal directly to the token scores that form the map: AUROC 0.761 under the same features, bags and labels on that national-set pool, and 0.802 for the swept head with 480 m of

neighborhood context on the expanded pool of Fig. 2. Because Mexican municipal grades are population-weighted aggregates of tract grades, the setting is label proportions with *known* instance weights; weighting each token by its tract’s census population is a robustness variant that performs equivalently ($\rho = 0.232$ against 0.227 on validation; 0.182 against 0.196 on test).

The sweep also quantified a limitation of model selection under weak supervision. Across the fourteen configurations, bag-level error, the only selection signal weak supervision may legitimately use, has a rank correlation of -0.14 with within-municipality map quality and $+0.62$ with the map’s pooled AUROC, the wrong sign for a selection criterion (Fig. 3b). Fourteen configurations give these correlations a standard error near 0.27, so the estimate is coarse; what it supports is that selection on bag error is close to uninformative about the quantity that matters. Whether this extends beyond settings where the bag label is an average of instance states is open, but any pipeline whose product is the map is affected, computational pathology included, which is why the benchmark includes held-out instance labels.

4.3 External validity

We ran four independent comparisons to test whether the map measures deprivation or a pipeline artifact (Fig. 4). Against the leading downloadable product, Meta’s Relative Wealth Index [3], scored per tract on an identical universe with a city-clustered bootstrap of the paired difference (2,958 shared AGEb), the map wins in 18 of 22 municipalities: $\Delta\rho = +0.22 [+0.08, +0.29]$; on the test cities the difference is $+0.13 [-0.07, +0.26]$, directionally consistent and not individually significant over 22 municipalities (Fig. 4a). A land-cover-plus-NDVI head with no learned representation reaches $\rho = 0.150$ on validation and 0.219 on test, where it is statistically indistinguishable from the full model (paired $\Delta\rho = -0.02 [-0.18, +0.10]$, the full model winning 16 of 29 municipalities). The learned features’ advantage within municipalities is therefore split-dependent; what holds on both splits is their edge on the bag task (bag error 0.095 against 0.110) and, on test, on high-deprivation detection (AUROC 0.821 against 0.781). Much of the within-municipality ordering is captured by land cover alone. This is consistent with the exchange rate, which measures what the published supervision recovers for a fixed feature set, whichever encoder produces it. Against an index from an institution whose data never entered training (CONAPO’s continuous urban marginalization index, built from a different indicator set over the same census), the map correlates at $\rho = 0.361 \pm 0.073$ within municipalities on validation and 0.257 ± 0.146 on test, in both cases at least as high as against its own five-level training construct, consistent with the continuous index resolving ties that five levels cannot (Fig. 4b). In simulated geographic targeting with fractional funding of the marginal tract, allocating by the map instead of the municipal aggregate closes 26–43% of the gap to census-guided allocation on validation at budgets of 10% and above, and 31–66% on the test cities with no crossover there (Fig. 4c). At the tightest validation budget (5%), where errors at the top of the ranking are most costly, the map loses to the aggregate. The map is also not an administrative artifact. Crossing a municipal border where deprivation does not change moves it by $+0.019 [-0.002, +0.155]$, and tract borders inside municipalities are crossed smoothly ($+0.001$). An earlier $+0.14$ border jump traced to bag-limited neighborhood adjacency at inference and disappeared after the adjacency was computed city-wide. The test cities contain too few same-grade cross-border pairs to repeat

215 this control there.

216 The sharper version of the question is what a practitioner would get for free. Four global
217 products were scored on the identical universe under the identical protocol (Fig. 5a, Supplemen-
218 tary Table 4). Against GHSL residential population the map wins clearly, $\Delta\rho = +0.29$ $[-0.00,$
219 $+0.50]$ on validation and $+0.19$ $[+0.05, +0.29]$ on test, the only one of the four that separates on
220 test. Against GHSL built-up surface ($+0.15$ $[-0.10, +0.30]$; $+0.06$ $[-0.10, +0.16]$) and GHSL
221 building height ($+0.13$ $[-0.10, +0.29]$; $+0.11$ $[-0.06, +0.23]$) the map leads on both splits and
222 wins in 16 of 22 municipalities on test, with intervals that include zero throughout. Against
223 nighttime lights the two are tied and the sign flips between splits: $+0.04$ $[-0.11, +0.15]$ on
224 validation, -0.04 $[-0.25, +0.10]$ on test. A 1 km light product, free and requiring no model,
225 orders tracts inside a Mexican municipality about as well as the map does. That comparison is
226 handicapped in the light product’s disfavor by resolution, since 63% of test tracts are smaller
227 than one of its cells and take the value at an interior point, and it is reported at face value
228 regardless: within the precision fourteen cities allow, we cannot show the map beats nighttime
229 lights.

230 The pooled comparison points the other way, and in doing so repeats the dissociation of the
231 previous section on external ground. For detecting high and very-high grade tracts across the
232 whole test set, all four products land at or above the map: built-up surface reaches AUROC
233 0.908, building height 0.886, population 0.855 and nighttime lights 0.831, against the map’s
234 0.829 (Fig. 5b). The products carry more of the national gradient, which is the easy half of the
235 problem and the half gridded global layers are built for. What the supervision buys, where it
236 buys anything, is the ordering inside a municipality, the half a targeting decision needs and the
237 half no global product was built to deliver.

238 4.4 Zero-shot transfer to Bogotá and Rio de Janeiro

239 Applied unchanged to Bogotá, the model orders Colombia’s block-level socioeconomic strata at
240 $\rho = 0.318$ (AUROC 0.716 for the deprived strata 1–2 over 14,132 tokens joined to blocks; Fig. 4d).
241 Applied to Rio de Janeiro, it detects IBGE’s mapped informal settlements barely above chance
242 (AUROC 0.557 among built-up tokens; Fig. 4d). The two targets differ in construct: Colombian
243 strata measure the same graded deprivation ordering as the GRS over the whole urban fabric,
244 while the AGSN mask delimits informality, a partly orthogonal notion. The model learned a
245 deprivation gradient from the aggregates, and that gradient transfers across countries; informality
246 detection does not follow from it.

247 4.5 Uncertainty, and the cities where the map fails

248 A map offered for targeting has to say where it is unsure. The ensemble of three seeds supplies the
249 obvious candidate, and it does not work. Seed disagreement ranks the tract errors at $\rho = 0.084$
250 $[0.002, 0.140]$ within municipalities on validation and 0.044 $[-0.011, 0.084]$ on test: a signal too
251 weak to act on, and on test indistinguishable from none. Acting on it anyway buys nothing.
252 Setting aside the least confident tracts inside each municipality, over a panel of municipalities
253 held fixed at every depth so attrition cannot manufacture a trend, leaves map quality flat: 0.390
254 to 0.369 from no discard to half discarded on validation, 0.317 to 0.327 on test, every point

inside every other point’s interval. (These are tract-mean correlations, higher than the token-level figures quoted above because averaging tokens to tracts removes noise; the comparison here is internal, so the estimator only has to be consistent with itself.) Ensembling over seeds does not produce an uncertainty layer for this problem, and we report no confidence raster.

What can be attached is an interval. Split conformal calibrated on the validation cities and evaluated on the test cities [17] gives, at 90% nominal coverage, a half-width of 1.29 grades under the marginal rule, and that rule undercovers: 0.869 overall and below nominal in 10 of the 14 test cities, worst at La Paz (0.768). Pricing in that cities are not exchangeable with one another, by taking each calibration city’s own quantile and then the 90th percentile across cities, restores coverage to 0.953 with only Madera below nominal, at the cost of a half-width of 1.68 grades (Fig. 5c). On a five-level scale that interval spans most of the range. The honest statement is that the map orders neighborhoods and cannot commit to a grade, and that any interval which ignores between-city variation will be too narrow.

The failures are concentrated and can be named. Ranking the fourteen test cities by the within-municipality correlation their map achieves, three sit at or below zero: Guerrero, Chihuahua (−0.51 over 21 tracts), Ciudad del Carmen, Campeche (−0.27 over 54), and Escárcega, Campeche (−0.00 over 24). All three are small, and the top of the ranking is not: Iguala de la Independencia (+0.64 over 133 tracts), La Paz (+0.53 over 276) and Ixtapaluca (+0.47 over 463). The pattern in the ranking is tract count rather than deprivation share, which is the modifiable-areal-unit effect of the Limits section arriving as a per-city failure mode: a municipality carved into two dozen tracts offers too few ranks for the correlation to stabilize, and the map’s ordering inside it should not be used. Seed spread does not flag these cities either; Guerrero’s median spread (0.101) sits at the middle of the fourteen.

4.6 Limits

The upper bound itself is low: even with full tract labels and neighborhood context, 160 m tokens support $\rho = 0.251$ within municipalities on validation and 0.233 on test: sufficient to order neighborhoods, too coarse to resolve individual blocks. Agreement depends strongly on tract size: $\rho = 0.26$ for tracts covered by 1–3 tokens, rising to 0.43 at 20–49 tokens, and collapsing to 0.13 in the two municipalities dominated by very large tracts; the test split reproduces the pattern noisily, peaking at 0.38 and collapsing in its single large-tract municipality. This is the modifiable areal unit problem, quantified here by tract size (Supplementary Table 2). Ecological caution applies throughout: the model recovers where deprived tracts are, and tract grades are themselves aggregates over households; neither the labels nor the maps identify deprived households, and the targeting simulation counts the population of deprived tracts; deprived individuals themselves are unobserved.

5 Discussion

Prior applications of this technology predict aggregates across space; the exchange rate measures how much sub-municipal detail can be recovered from statistics published at a given resolution, which is the question a statistics office would ask. In operational terms: the national supply of

municipal aggregates recovers 84–90% of what these features support, coarser publication reduces recovery in proportion, and sensor content rather than resolution is the binding property. The targeting experiment expresses the result as people reached, including the one budget where the map underperforms the aggregate. Three negative results matter for similar efforts: attention-based pooling fails for a structural reason, selection on aggregate error is close to uninformative about map quality, and informality detection does not follow from a deprivation gradient. All three will recur in other weakly supervised mapping work.

The comparison against free global grids sets the operational recommendation more honestly than an accuracy table would. On the pooled task every downloadable product outperforms the map, and within municipalities only the population grid separates from it on test, with nighttime lights matching it on both splits. So where the question is which municipalities are poor, a global layer answers it today at no cost; where the question is which neighborhoods inside a municipality are poorest, the aggregate-trained map is the better instrument, by a margin smaller than the enthusiasm in this literature would suggest. The missing uncertainty layer sharpens the same point. Seeds disagree where the map is right about as often as where it is wrong, and the interval that does hold its coverage across cities spans 1.7 of the five grades. What this method supports publishing is a ranking with the small municipalities named and excluded; a graded raster would overstate it.

Mexico’s dual-level publication made the calibration possible; the measured fractions can be extrapolated to countries that publish only aggregates, and the Bogotá transfer is a first test of that extrapolation. Extending the dual calibration to Colombia’s own strata, holding out the block-level strata and building coarse aggregates from them, is the natural next measurement, and every number here is reproducible from the released frozen features.

6 Conclusions

Municipal aggregates, the level at which most countries actually publish social statistics, carry most of the neighborhood-scale ordering that frozen satellite features support: 84% of the fully supervised upper bound on unseen test cities, 90% on validation. The upper bound itself is low ($\rho = 0.23$ within municipalities at 160 m), which bounds every claim in this paper and, we would argue, every claim in this setting built on free decametric imagery. Within that bound, the supervision-efficiency analysis is operational for statistics offices: a few hundred aggregates recover most of what the full national supply does, coarser publication forfeits recovery in proportion, and sensor content matters more than resolution. Two methodological results travel beyond the application: attention-based multiple-instance learning produces chance-level maps whenever the bag task is solvable from the feature mean, and model selection on bag-level error is close to uninformative about map quality, which argues for benchmarks with held-out instance labels wherever maps are the product. A third caution is operational: measured against the free global grids a practitioner already has, the map’s advantage is confined to the within-municipality ordering and is resolvable on fourteen test cities only against gridded population, while ensembling supplies no uncertainty layer and an honest grade interval spans 1.7 of the five levels. All data, frozen features, labels, splits and code are released.

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Data availability

The frozen benchmark (foundation-model vectors for every bag, municipal and tract labels, splits, and the evaluation protocol) will be deposited on Zenodo with a DOI at publication. All source data are public: Copernicus Sentinel via Microsoft Planetary Computer, CONEVAL, CONAPO, INEGI, ESA WorldCover, Meta RWI (HDX), the GHSL R2023A grids from the JRC data portal, the harmonized nighttime-light layer from Figshare, IBGE, and Bogotá’s open-data portal.

Code availability

The full pipeline (MIT licence) is available at the project repository, pinned by `uv.lock`. Each figure maps to one driver script under `herramientas/`: `figura_diseno.py` for Fig. 1, `figura_curva.py` for Fig. 2, `figura_disociacion.py` for Fig. 3, `figura_validez.py` for Fig. 4 and `figura_incumbentes.py` for Fig. 5. Each recomputes the quantities it draws from the per-seed artifacts and halts if any of them departs from the canonical results file, so a figure and the text cannot disagree silently.

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CRedit authorship contribution statement

Dagoberto Pulido Arias: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing, Visualization.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have influenced this work.

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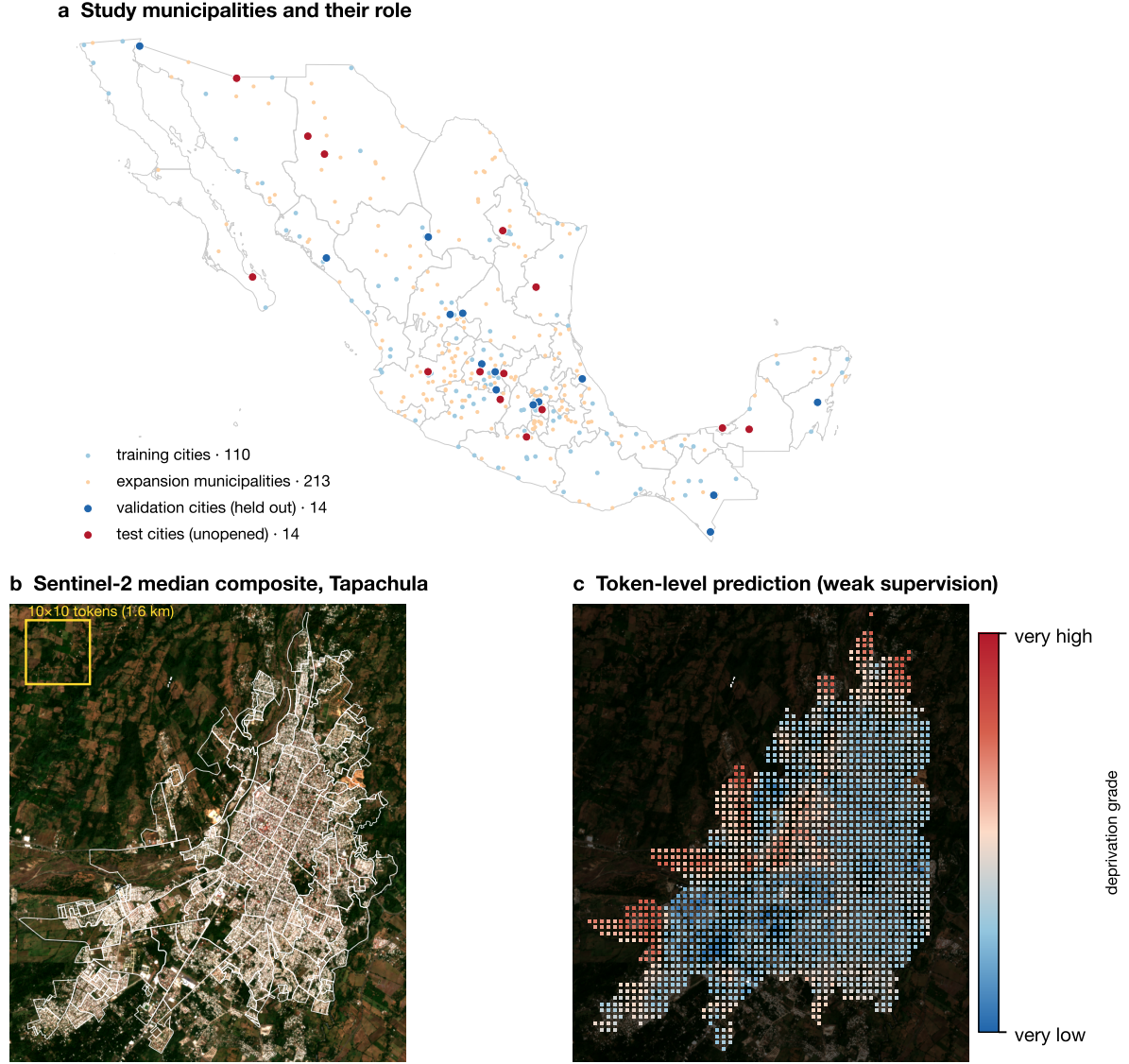


Figure 1: **Study design.** **a**, The 351 study units: 110 training cities, the 213-municipality expansion toward the deprived half of urban Mexico (municipalities overlapping held-out ground excluded), 14 validation and 14 unopened test cities. A city box is a conurbation and typically yields several municipal bags, which is how 351 units produce 771 bags. **b**, What the model consumes: the 2020 Sentinel-2 median composite of Tapachula (a validation city) with AGEb boundaries overlaid; tokens of 160 m tile the box. **c**, What it produces: the token-level deprivation score from municipal aggregates alone. Tract ground truth for the same city is shown in Supplementary Fig. 1; Tapachula sits at the validation median ($\rho = 0.47$), chosen as a median-performing, representative example.

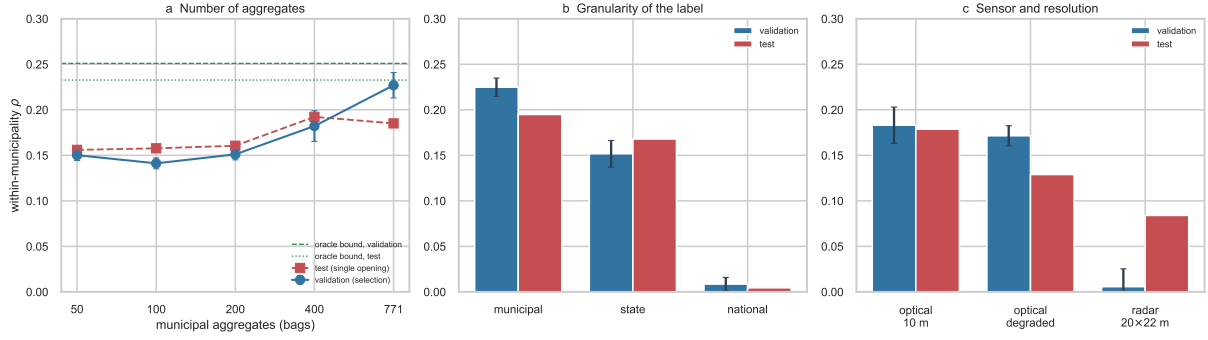


Figure 2: **The exchange rate between aggregates and maps.** All panels: mean over three seeds against held-out tract grades; blue, the 14 validation cities used for selection (error bars, s.d. over seeds); red, the 14 test cities scored once with frozen models and drawn without a spread; dashed and dotted lines, the fully supervised oracle upper bound on each split ($\rho = 0.251$ validation, 0.233 test). **a**, Agreement as a function of how many municipal aggregates supervise training (nested, grade-stratified sampling). **b**, The same 771 bags relabeled with coarser aggregates. **c**, Single-sensor configurations: degrading optical resolution to the radar’s 20 m grain costs little on validation and a real but secondary amount on test; radar sits far below both.

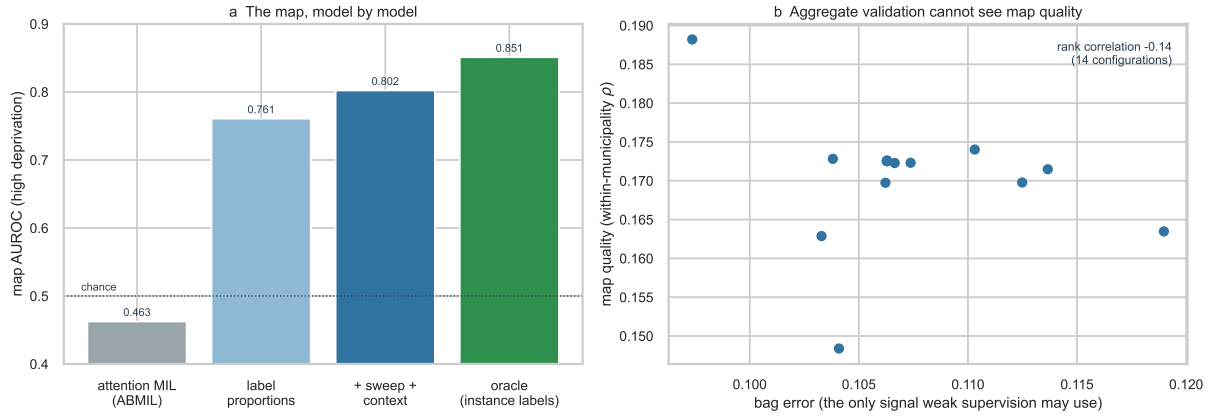


Figure 3: **Bag prediction versus localization.** **a**, Map quality by model family on identical features and labels; the first three bars are measured on the national-set pool, the swept bar on the expanded pool of Fig. 2. Attention pooling solves the bag task from the feature mean and receives no localization gradient; predicting instances and averaging predictions makes the map the mechanism. **b**, Across the fourteen sweep configurations (grouped training folds of the national set), the only criterion weak supervision may monitor is nearly uncorrelated with within-municipality map quality, and positively correlated ($+0.62$) with the pooled AUROC it would seem to proxy.

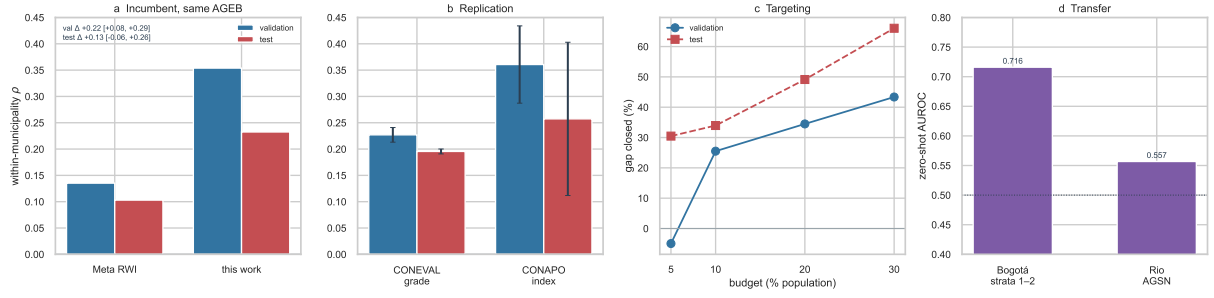


Figure 4: **External validity.** Blue, validation; red, test (single opening); purple in **d**, which belongs to neither split. **a**, Within-municipality agreement on the identical AGEB universe reached by the incumbent’s grid; annotation, city-clustered bootstraps of the paired difference per split. **b**, Replication against an independent institution’s continuous index, beside the map’s agreement with the construct it was trained toward (error bars: s.d. over seeds on the CONEVAL bars, city-clustered bootstrap half-widths on the CONAPO bars). **c**, Share of the aggregate-to-census targeting gap closed at each budget, people pooled per split; validation loses at the tightest budget, test does not. **d**, Zero-shot transfer: the Mexican model, unchanged and unadapted, ordering Bogotá’s block strata (AUROC for strata 1–2; $n = 14,132$ tokens) and detecting Rio’s mapped informal settlements among built-up tokens ($n = 34,331$). The model transfers to a graded deprivation target and fails on informality, a different construct.

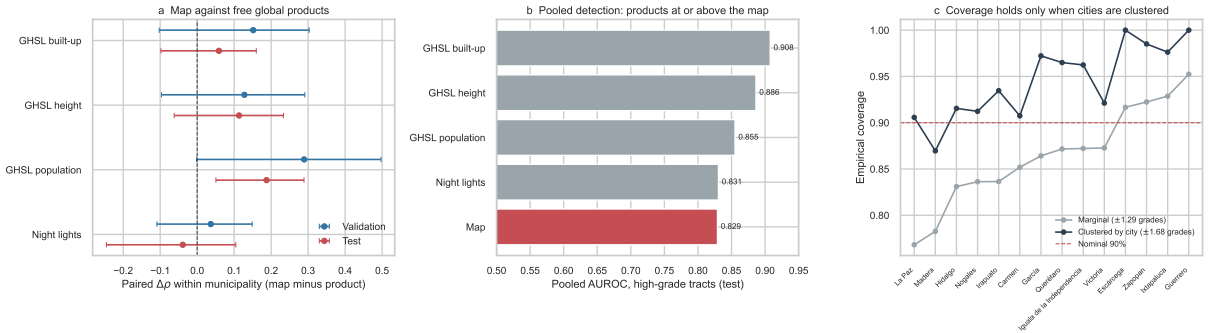


Figure 5: **Free products, and the confidence the map can carry.** **a**, Paired within-municipality difference between the map and each downloadable global product on the identical tract universe, blue validation and red test, with city-clustered bootstrap intervals; only gridded population separates from the map on test, and nighttime lights straddle zero on both splits. **b**, Pooled detection of high and very-high grade tracts across the test cities: all four products land at or above the map, which is the bag-versus-map dissociation of Fig. 3 arriving on external ground. **c**, Conformal coverage per test city at 90% nominal, calibrated on the validation cities. The marginal rule falls below nominal in ten of fourteen cities; the clustered rule restores coverage at a half-width of 1.68 of the five grades.